```
import pandas as pd

url=
"https://raw.githubusercontent.com/datasciencedojo/datasets/master/
titanic.csv"
titanic_df = pd.read_csv(url)
print(titanic_df.head())
```

```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age
SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0
1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0
1
2                             Heikkinen, Miss. Laina  female  26.0
0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0
1
4                           Allen, Mr. William Henry    male  35.0
0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

```
import seaborn as sns
import pandas as pd

# in python sns there are some dataset already existing, we can load
them using sns.load_dataset('dataset_name')
df= sns.load_dataset('titanic')
print(df.head())
```

```
   survived  pclass     sex   age  sibsp  parch     fare embarked
class  \
0         0       3    male  22.0      1      0   7.2500        S
Third
1         1       1  female  38.0      1      0  71.2833        C
First
2         1       3  female  26.0      0      0   7.9250        S
```

```
Third
3         1       1  female  35.0      1      0  53.1000        S
First
4         0       3    male  35.0      0      0   8.0500        S
Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True
```

```
#1 What is the size of the dataset?

print("Dataset shape: ", df.shape)
print("Number of row: ", df.shape[0])
print("Number of column: ", df.shape[1])

print("Dimention of the dataset: ", df.ndim)
```

```
Dataset shape:  (891, 15)
Number of row:  891
Number of column:  15
Dimention of the dataset:  2
```

```
#2 How does the dataset look like?
print(df.head())
print(df.tail(10))
```

```
   survived  pclass     sex   age  sibsp  parch     fare embarked
class  \
0         0       3    male  22.0      1      0   7.2500        S
Third
1         1       1  female  38.0      1      0  71.2833        C
First
2         1       3  female  26.0      0      0   7.9250        S
Third
3         1       1  female  35.0      1      0  53.1000        S
First
4         0       3    male  35.0      0      0   8.0500        S
Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True
     survived  pclass     sex   age  sibsp  parch     fare embarked
class  \
```

```
881         0       3    male  33.0      0      0   7.8958         S
Third
882         0       3  female  22.0      0      0  10.5167         S
Third
883         0       2    male  28.0      0      0  10.5000         S
Second
884         0       3    male  25.0      0      0   7.0500         S
Third
885         0       3  female  39.0      0      5  29.1250         Q
Third
886         0       2    male  27.0      0      0  13.0000         S
Second
887         1       1  female  19.0      0      0  30.0000         S
First
888         0       3  female   NaN      1      2  23.4500         S
Third
889         1       1    male  26.0      0      0  30.0000         C
First
890         0       3    male  32.0      0      0   7.7500         Q
Third

       who  adult_male deck  embark_town alive  alone
881    man        True  NaN  Southampton    no   True
882  woman       False  NaN  Southampton    no   True
883    man        True  NaN  Southampton    no   True
884    man        True  NaN  Southampton    no   True
885  woman       False  NaN   Queenstown    no  False
886    man        True  NaN  Southampton    no   True
887  woman       False    B  Southampton   yes   True
888  woman       False  NaN  Southampton    no  False
889    man        True    C    Cherbourg   yes   True
890    man        True  NaN   Queenstown    no   True

#print any random row
df.sample()
df.sample(10)
df.sample(frac=0.1)

     survived  pclass     sex   age  sibsp  parch     fare embarked
class  \
7           0       3    male   2.0      3      1  21.0750        S
Third
656         0       3    male   NaN      0      0   7.8958        S
Third
780         1       3  female  13.0      0      0   7.2292        C
Third
551         0       2    male  27.0      0      0  26.0000        S
Second
815         0       1    male   NaN      0      0   0.0000        S
First
```

```
..        ...     ...     ...   ...    ...    ...      ...       ...
...
473         1       2  female  23.0      0      0  13.7917         C
Second
668         0       3    male  43.0      0      0   8.0500         S
Third
150         0       2    male  51.0      0      0  12.5250         S
Second
108         0       3    male  38.0      0      0   7.8958         S
Third
837         0       3    male   NaN      0      0   8.0500         S
Third

       who  adult_male deck  embark_town alive  alone
7    child       False  NaN  Southampton    no  False
656    man        True  NaN  Southampton    no   True
780  child       False  NaN    Cherbourg   yes   True
551    man        True  NaN  Southampton    no   True
815    man        True    B  Southampton    no   True
..     ...         ...  ...          ...   ...    ...
473  woman       False    D    Cherbourg   yes   True
668    man        True  NaN  Southampton    no   True
150    man        True  NaN  Southampton    no   True
108    man        True  NaN  Southampton    no   True
837    man        True  NaN  Southampton    no   True

[89 rows x 15 columns]

df.tail(7)

     survived  pclass     sex   age  sibsp  parch    fare embarked
class  \
884         0       3    male  25.0      0      0   7.050         S
Third
885         0       3  female  39.0      0      5  29.125         Q
Third
886         0       2    male  27.0      0      0  13.000         S
Second
887         1       1  female  19.0      0      0  30.000         S
First
888         0       3  female   NaN      1      2  23.450         S
Third
889         1       1    male  26.0      0      0  30.000         C
First
890         0       3    male  32.0      0      0   7.750         Q
Third

       who  adult_male deck  embark_town alive  alone
884    man        True  NaN  Southampton    no   True
885  woman       False  NaN   Queenstown    no  False
```

```
886    man        True  NaN  Southampton    no   True
887  woman       False    B  Southampton   yes   True
888  woman       False  NaN  Southampton    no  False
889    man        True    C    Cherbourg   yes   True
890    man        True  NaN   Queenstown    no   True
```

```
#3 What is the data type of each column?

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype   
---  ------       --------------  -----   
 0   survived     891 non-null    int64   
 1   pclass       891 non-null    int64   
 2   sex          891 non-null    object  
 3   age          714 non-null    float64 
 4   sibsp        891 non-null    int64   
 5   parch        891 non-null    int64   
 6   fare         891 non-null    float64 
 7   embarked     889 non-null    object  
 8   class        891 non-null    category
 9   who          891 non-null    object  
 10  adult_male   891 non-null    bool    
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object  
 13  alive        891 non-null    object  
 14  alone        891 non-null    bool    
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

```
Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
       'embarked', 'class', 'who', 'adult_male', 'deck',
'embark_town',
       'alive', 'alone'],
      dtype='object')

display(df.columns)
```

```
Index(['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare',
       'embarked', 'class', 'who', 'adult_male', 'deck',
'embark_town',
       'alive', 'alone'],
      dtype='object')

#4 Are there any missing values in the dataset?

df.isnull().sum()
```

```
survived          0
pclass            0
sex               0
age             177
sibsp             0
parch             0
fare              0
embarked          2
class             0
who               0
adult_male        0
deck            688
embark_town       2
alive             0
alone             0
dtype: int64
```

```
#5 How does the dataset look mathematically? (mean, median, mode,
standard deviation, etc.)

display(df.describe())
```

```
         survived      pclass         age       sibsp       parch
fare
count  891.000000  891.000000  714.000000  891.000000  891.000000
891.000000
mean     0.383838    2.308642   29.699118    0.523008    0.381594
32.204208
std      0.486592    0.836071   14.526497    1.102743    0.806057
49.693429
min      0.000000    1.000000    0.420000    0.000000    0.000000
0.000000
25%      0.000000    2.000000   20.125000    0.000000    0.000000
7.910400
50%      0.000000    3.000000   28.000000    0.000000    0.000000
14.454200
75%      1.000000    3.000000   38.000000    1.000000    0.000000
31.000000
max      1.000000    3.000000   80.000000    8.000000    6.000000
512.329200
```

```
print(df.describe().loc['mean', 'age'])
print(df['age'].mean())
```

```
29.69911764705882
29.69911764705882
```

```
#6 Are there any duplicate values?
print(df.duplicated().sum())
```

```
107
```

```
#remove all duplicated values

df=df.drop_duplicates()
df.reset_index(drop=True, inplace=True)
df.duplicated().sum()
```

```
np.int64(0)
```

```
#7 How is the correlation among columns?

#df.select_dtypes(include=['float64', 'int64']).corr()
#non column correlation
df.select_dtypes(include=['float64', 'int64']).corrwith(df['age'])
```

```
survived   -0.086548
pclass     -0.369361
age         1.000000
sibsp      -0.315116
parch      -0.195036
fare        0.092707
dtype: float64
```

```
import numpy as np
df.select_dtypes(include=[np.number]).corr()['survived']
```

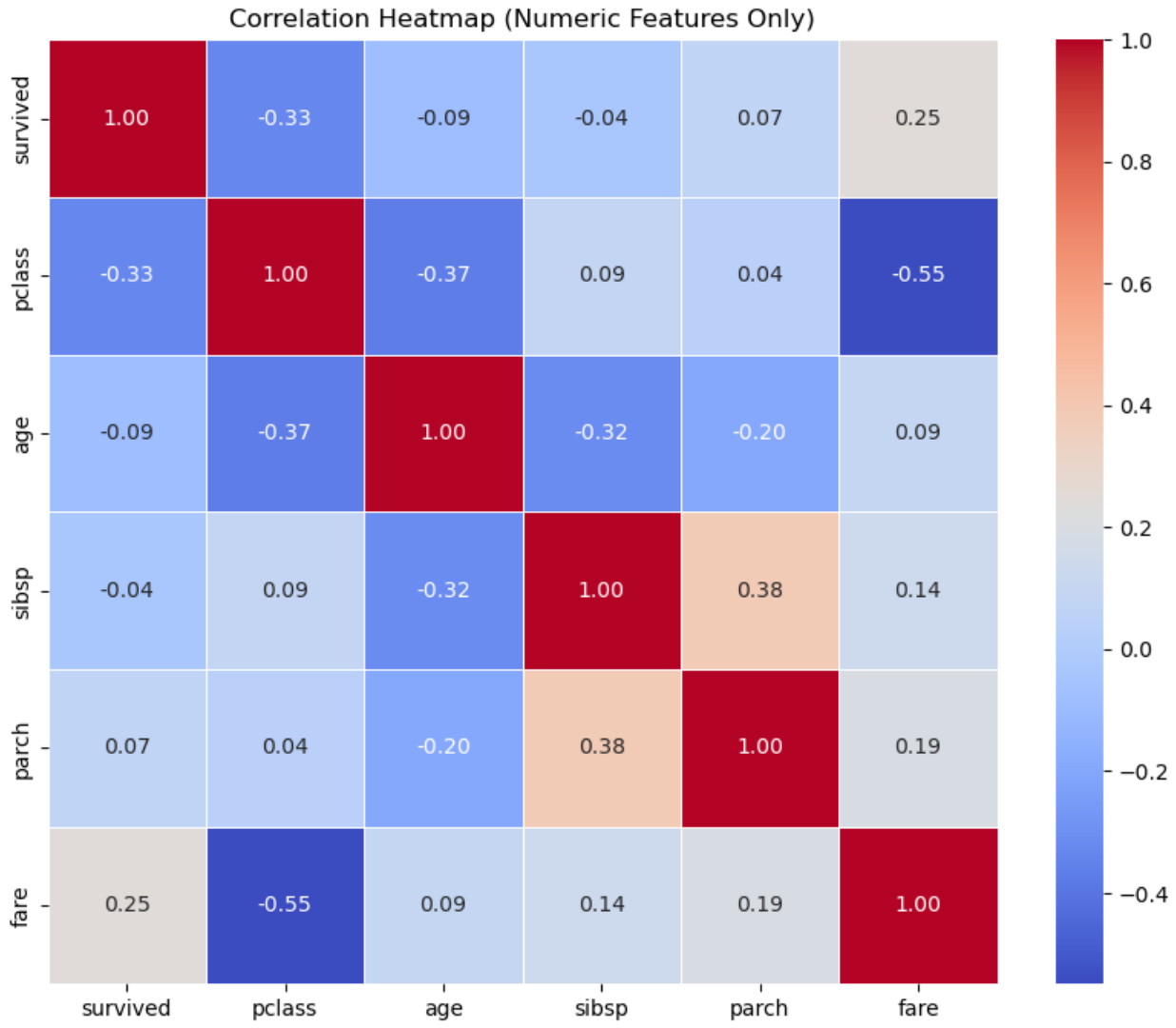
```
survived    1.000000
pclass     -0.332658
age        -0.086548
sibsp      -0.036589
parch       0.070307
fare        0.246769
Name: survived, dtype: float64
```

```
import matplotlib.pyplot as plt

numeric_df= df.select_dtypes(include='number')
correlations= numeric_df.corr(method='pearson')

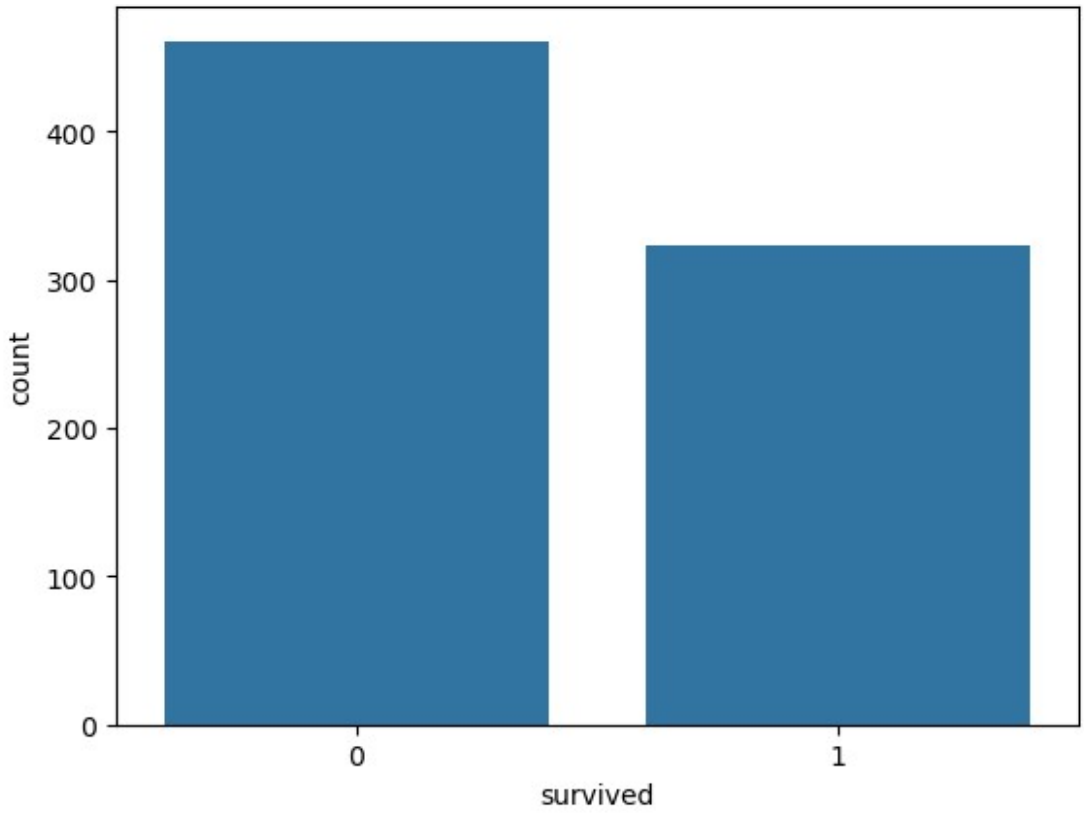
plt.figure(figsize=(10,8))
sns.heatmap(correlations, annot=True, fmt='.2f', cmap='coolwarm',
linewidths=0.5)

plt.title('Correlation Heatmap (Numeric Features Only)')
plt.show()
```




```
sns.countplot(df, x=df['survived'])
```

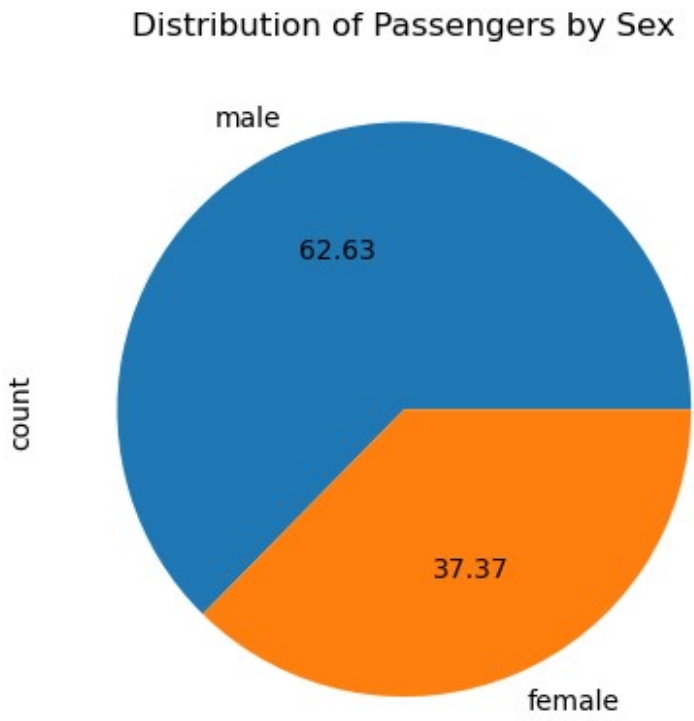
```
<Axes: xlabel='survived', ylabel='count'>
```



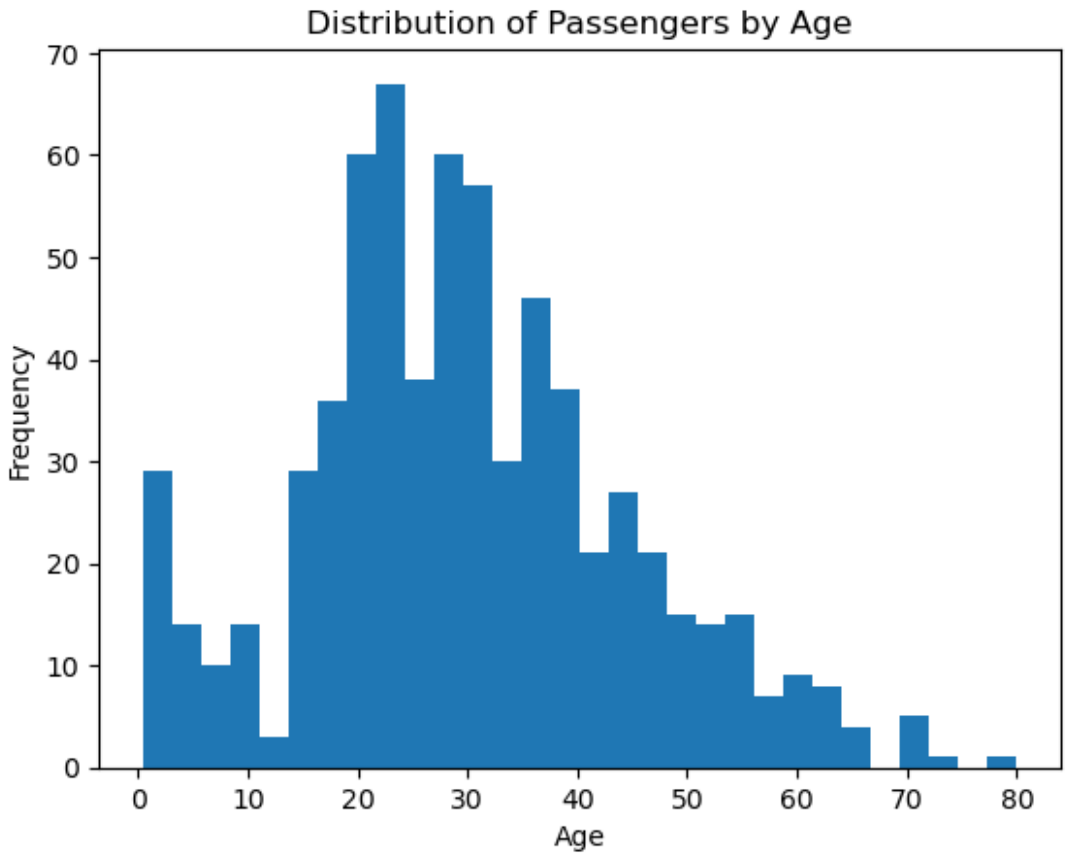

```
df['survived'].value_counts()
```

```
survived
0    461
1    323
Name: count, dtype: int64
```

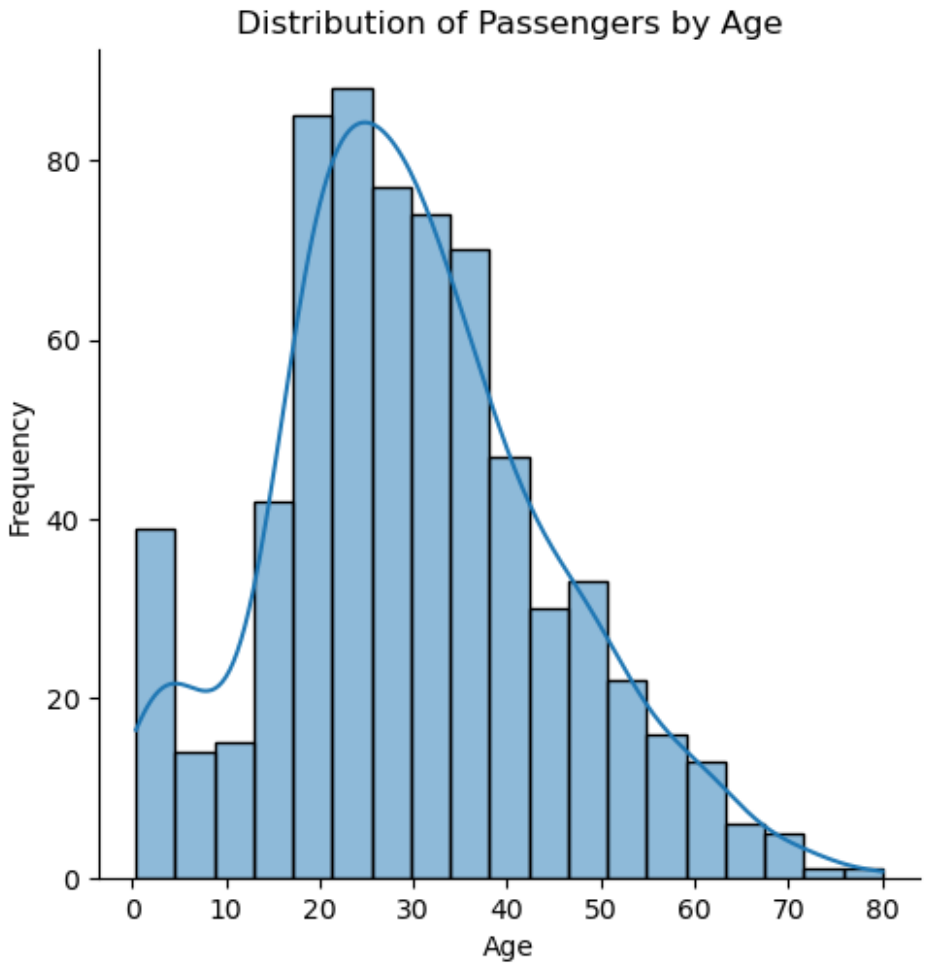
```
df['sex'].value_counts().plot(kind='pie', autopct='%.2f')
plt.title('Distribution of Passengers by Sex')
plt.show()
```




```
plt.hist(df['age'], bins=30)
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title('Distribution of Passengers by Age')
plt.show()
```

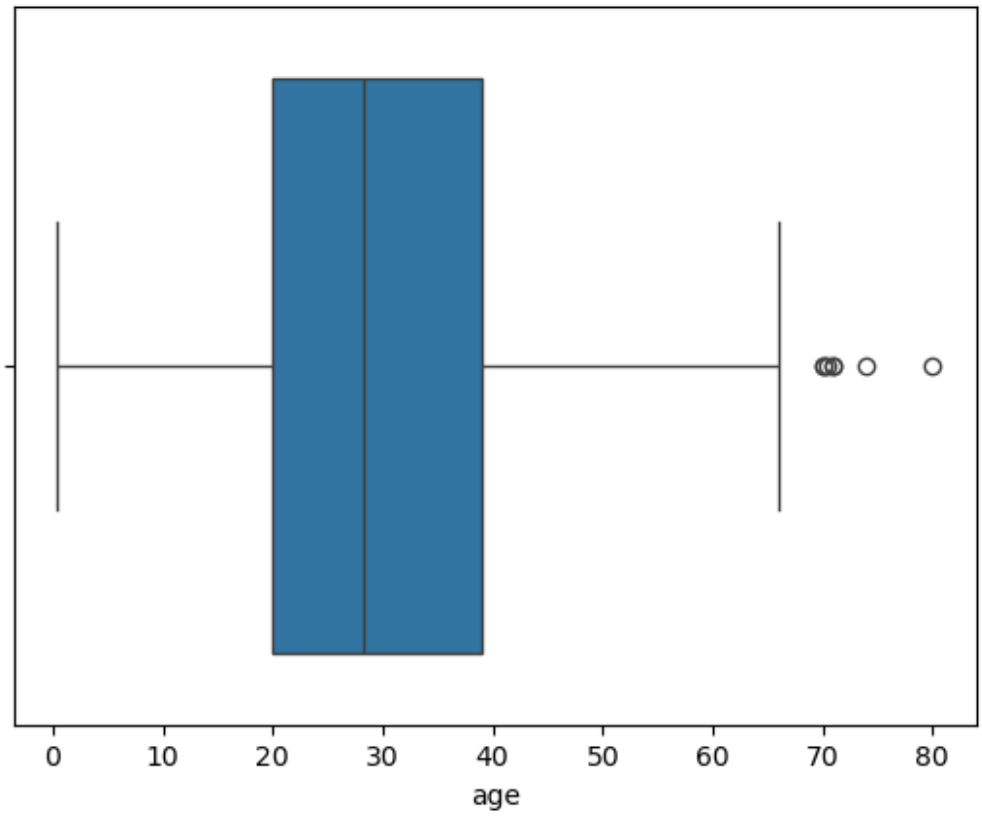



```
sns.displot(df['age'], kde=True)
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.title('Distribution of Passengers by Age')
plt.show()
```




```
sns.boxplot(x='age', data=df)
```

```
<Axes: xlabel='age'>
```

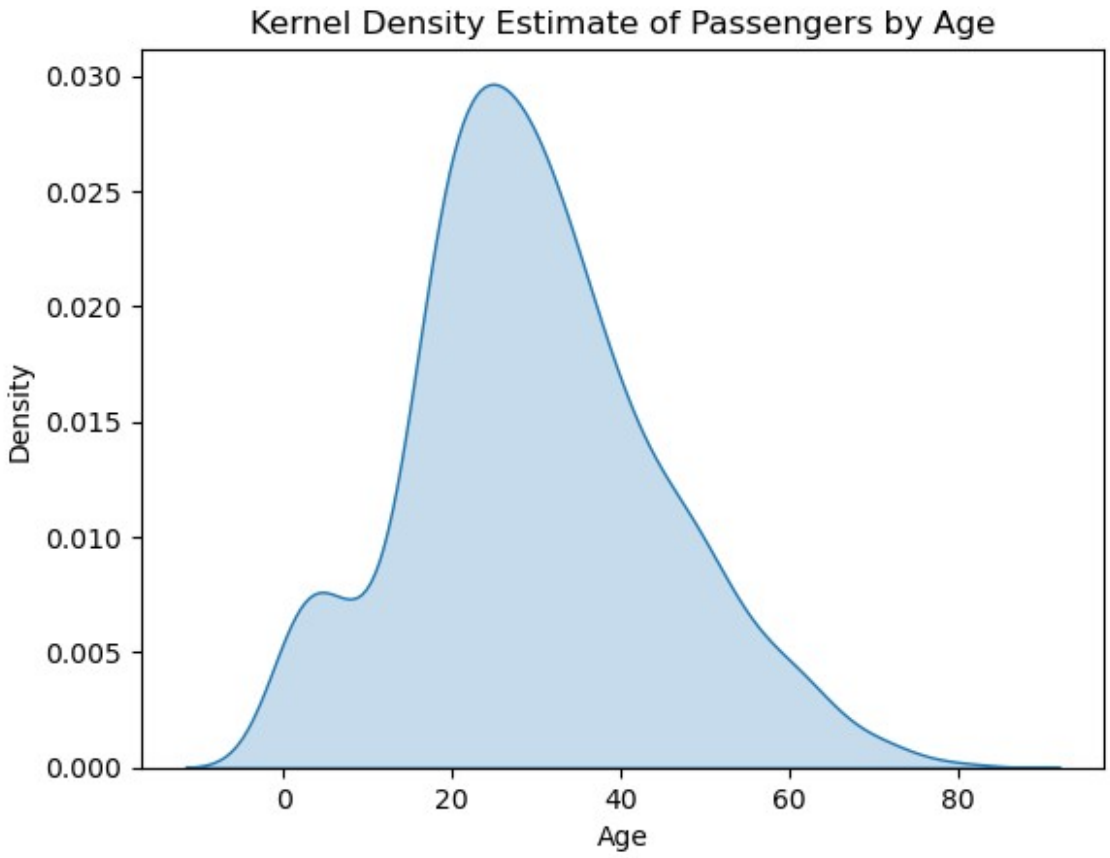



```
sns.kdeplot(df['age'], shade=True)
plt.xlabel('Age')
plt.ylabel('Density')
plt.title('Kernel Density Estimate of Passengers by Age')
plt.show()
```

```
C:\Users\mdfah\AppData\Local\Temp\ipykernel_38688\4119159821.py:1:
FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

  sns.kdeplot(df['age'], shade=True)
```




```
df['age'].skew()
```

```
np.float64(0.3674542095476674)
```